ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Antoine Lavoisier	Lavoisier.
2	**(b)** 4	Four atoms.
3	**(c)** negatively charged ion	Gains electrons.
4	**(b)** Joseph Proust	Proust.
5	**(d)** 8	Eight atoms.
6	**(b)** $NaCl$	1:1.
7	**(b)** equals the total mass of the products	Mass is neither created nor destroyed.
8	**(b)** 18 u	2 + 16.
9	**(d)** $CaCl_2$	Ca2+ and Cl-.
10	**(c)** 44 g	12 + 32 = 44.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass is neither created nor destroyed in a chemical reaction (1); Antoine Lavoisier (1).	1+1
12	Number of atoms in a molecule (1); 2, 3 and 4 (1).	1+1
13	By sharing of electron pairs between atoms (1); H_2 or Cl_2 (1).	1+1
14	A pure compound always contains elements in a fixed ratio by mass (1); Water: H : O = 1 : 8 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mass of reactants = 22.0 g (1); Mass of CO2 formed = 22.0 g (1); Mass before equals mass after (1).	1+1+1
16	Hard, brittle crystalline solids (1); High melting and boiling points (1); Conduct electricity when molten or dissolved (1).	1+1+1
17	Atom: smallest particle of an element (1); Molecule: group of atoms bonded (1); Atoms of noble gases exist alone; others usually as molecules (1).	1+1+1
	OR: Fraction of oxygen = 8/9 (1); Mass = 8/9 x 27 (1); = 24 g (1).	1+1+1
18	Low melting and boiling points (1); Poor conductors (1); Often soluble in organic solvents (1).	1+1+1
	OR: Oxygen/carbon = 32/12 (1); = 8 g (1); CO2 formed = 11 g (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take both solutions in a flask with a small tube (1); Weigh the set-up (1); Mix: white precipitate of BaSO4 forms (1); Weigh again: mass is the same (1); Conclusion: mass is conserved (1).	1+1+1+1+1
	OR: Oxygen = 8/3 x 18 = 48 g (1); Compound mass = 66 g (1); Compound is carbon dioxide (1); Ratio fixed by law of constant proportions (1); Conservation of mass verified (1).	1+1+1+1+1
20	Matter made of atoms (1); Atoms of an element identical (1); Atoms of different elements differ (1); Atoms combine in simple ratios (1); Atoms are neither created nor destroyed; later: atoms divisible, isotopes exist (1).	1+1+1+1+1
	OR: H_2O = 18 u (1); CH_4 = 16 u (1); C_2H_5OH = 46 u (1); HNO_3 = 63 u (1); Na_2CO_3 = 106 u (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.
- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.
- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).
- Calling $NaCl$ a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 44 u	12 + 32.
2	**(b)** Al_2O_3	Al3+ and O2-.
3	**(d)** 1 : 8	Fixed ratio.
4	**(d)** 17 u	14 + 3.
5	**(a)** MgO	Mg2+ and O2-.
6	**(c)** 1 g	1/9 of 9.
7	**(d)** 58.5 u	23 + 35.5.
8	**(b)** ionic bond	Ionic.
9	**(b)** Atoms are indivisible	Atoms contain electrons, protons and neutrons.
10	**(a)** 100 u	40 + 12 + 48.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	14 + 3 x 1 (1); = 17 u (1).	1+1
12	(a) Carbon monoxide (1); (b) Hydrogen sulphide (1).	1+1
13	All matter is made of tiny indivisible atoms (1); Atoms of an element are identical; atoms combine in simple whole-number ratios (1).	1+1
14	40 + 2 x 35.5 (1); = 111 u (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Na loses one electron to form Na^+ (1); Cl gains one electron to form Cl^- (1); Ions attract by electrostatic force: ionic bond (1).	1+1+1
16	Iron combines with oxygen and water from air (1); Total mass of iron and oxygen = rust (1); Mass is conserved in the whole system (1).	1+1+1
17	In solution ions are free to move (1); In the solid state ions are fixed in the lattice (1); Moving ions carry charge (1).	1+1+1
	OR: Each Cl atom has 7 valence electrons (1); Each shares one electron (1); A single covalent bond forms, completing the octet (1).	1+1+1
18	Atoms of each element have a fixed mass (1); Atoms combine in fixed whole-number ratios (1); Hence the mass ratio is fixed (1).	1+1+1
	OR: Molecular mass = 18 (1); 16/18 x 100 (1); = 88.9 per cent (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mg loses 2 electrons to form Mg^{2+} (1); Each Cl gains 1 electron to form Cl^- (1); $MgCl_2$ (1); Na loses 1 electron each (two Na) to O which gains 2 (1); Na_2O (1).	1+1+1+1+1
	OR: Conservation of mass showed atoms are not lost (1); Constant proportions showed fixed ratios (1); Dalton explained both using atoms (1); Atomic masses were derived (1); Basis of chemical formulae (1).	1+1+1+1+1
20	H_2: one shared pair, single bond (1); O_2: two shared pairs, double bond (1); N_2: three shared pairs, triple bond (1); Octets completed (1); Molecules of elements (1).	1+1+1+1+1
	OR: Na gives an electron to Cl (1); Ions arranged in a lattice (1); Strong electrostatic forces (1); Brittle: displaced layers repel (1); Melting needs much energy (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.

- Forgetting the bracket when a polyatomic ion occurs more than once: $\mathrm{Ca(OH)_2}$, not $\mathrm{CaOH_2}$.
- Mixing atomicity with number of molecules: $3\,\mathrm{O_2}$ means three molecules, six atoms; $\mathrm{O_3}$ is one molecule of ozone.
- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $\mathrm{(NH_4)_2SO_4}$: N counted twice, H eight times).
- Calling NaCl a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** covalent bond	Covalent.
2	**(b)** small whole-number ratios	Small whole numbers.
3	**(c)** 95 u	24 + 71.
4	**(a)** $NaCl$	Metal and non-metal.
5	**(b)** two or more atoms chemically bonded together that can exist independently	Group of atoms.
6	**(c)** a charged atom or group of atoms	Has a net charge.
7	**(c)** CH_4	Non-metals sharing electrons.
8	**(a)** 3	O_3.
9	**(a)** positively charged ion	Loses electrons.
10	**(d)** conduct electricity	Free ions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(a) Na_2O (1); (b) CaO (1).	1+1
12	Group of atoms chemically combined, existing independently (1); O_2 and H_2O (1).	1+1
13	Charged atom or group of atoms (1); Cation is positive; anion is negative (1).	1+1
14	Components in a mixture are in variable ratio (1); They are not chemically combined (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(a) $CaCl_2$ (1); (b) Al_2O_3 (1); (c) Mg_3N_2 (1).	1+1+1
16	Sodium, magnesium, chloride (1); Oxide, hydroxide (1); Ammonium (1).	1+1+1
17	Group of atoms carrying a charge (1); SO_4^{2-}, NO_3^- (1); NH_4^+ (1).	1+1+1
	OR: CO_2: 12 + 32 = 44 u (1); H_2SO_4: 2 + 32 + 64 (1); = 98 u (1).	1+1+1
18	(a) Na_2SO_4 (1); (b) $Ca(OH)_2$ (1); Criss-cross of charges (1).	1+1+1
	OR: Helium has a complete outer shell, stable (1); Oxygen atoms share electrons to complete the octet (1); Hence O2 molecule (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Bond: transfer vs sharing (1); State: solids vs liquids/gases/solids (1); Melting point: high vs low (1); Conductivity: conduct when molten vs poor (1); Solubility: water vs organic solvents (1).	1+1+1+1+1
	OR: Atom: Na, He (1); Molecule: O_2, H_2O (1); Ion: Na^+ (1); Formula unit: NaCl (1); Atomicity: number of atoms; valency: combining capacity (1).	1+1+1+1+1
20	Write symbols with charges (1); Criss-cross the charges (1); Use brackets for polyatomic ions (1); Simplify the ratio (1); Examples applied (1).	1+1+1+1+1
	OR: Mass of CaCO3 = 100 u (1); Percentage of Ca = 40 per cent (1); Mass in 250 g = 100 g (1); Oxygen = 48 per cent (1); Carbon = 12 per cent (1).	1+1+1+1+1

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